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1. Motivation and goals

While significant research has focused on developing diverse methodologies for data traffic collection and processing, the assessment of data quality remains comparatively overlooked. The aim of this work is to design and implement a method to:

(i) **quantify and estimate the errors** derived from traffic counts of radar sensors as probabilistic mismatch w.r.t. a baseline yielded by camera detections;

(ii) **reconstruct a plausible temporal association** between the traffic counts of radar sensors and camera detections.



2. A Bayesian error estimation framework

Objectives:

- (i) compute the probability *a posteriori* for the camera to show the passage of a vehicle given that the radar has detected no vehicle (or vice-versa);
- (ii) find the most probable matching in terms of time instant between camera and radar detections, considering the four vehicular categories in such an association.

Solution:

Hidden process: vehicle transit $X(t) \in \{0,1\}$, $t \in \mathbb{Z}$

Observed process: radar detections $Y(t) \in \{0,1\}$, $t \in \mathbb{Z}$

- camera detections: $x(t)$ can be treated as outcomes of $X(t) \Rightarrow$ oracle $\{x(t)\}$
- radar detections $y(t)$ are the outcomes of $Y(t) \Rightarrow$ measurements $\{y(t)\}$

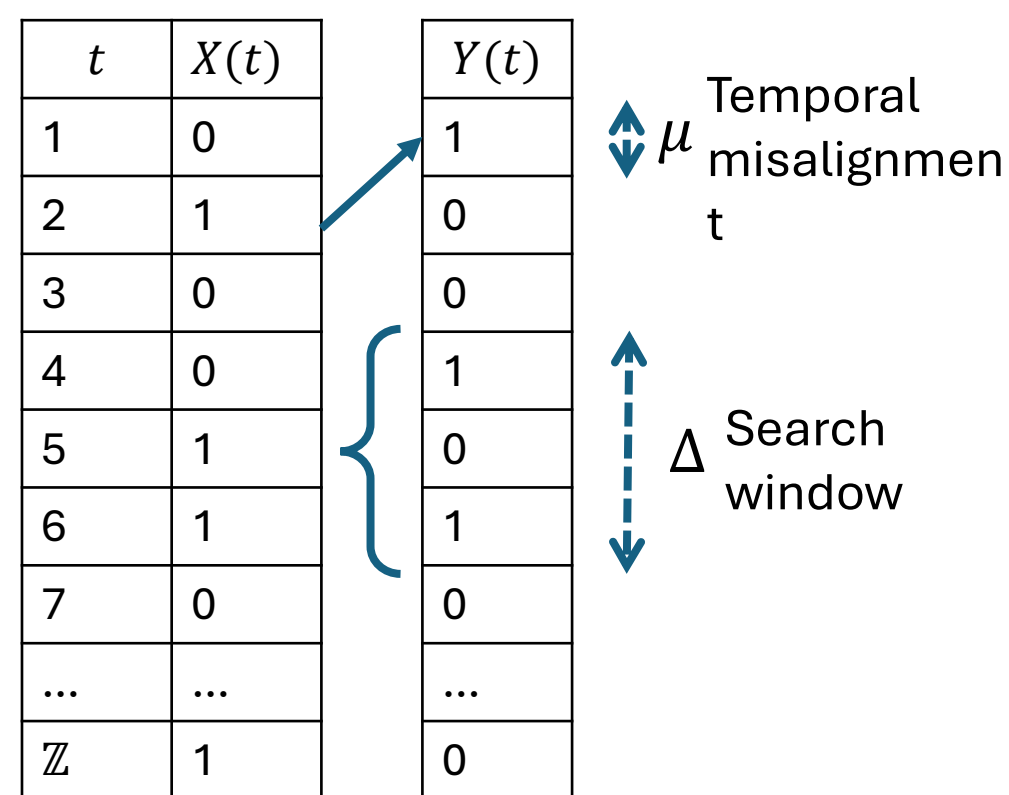
Main assumption: (X, Y) can be formalized as a Higher-order Hidden Markov Model (HHMM). Let $T \in \mathbb{N} \setminus \{0\}$ and $\Delta \in \mathbb{N}$ such that $\Delta \leq \lfloor T/2 \rfloor$.

Define the set

$$[\Delta] = \{-\Delta, \Delta + 1, \dots, 0, \dots, \Delta - 1, \Delta\}.$$

Then

$$\Pr[Y(t) = y(t) \mid \{X(\tau) = x(\tau)\}_{\tau \in \mathbb{Z}}] = \Pr[Y(t) = y(t) \mid \{X(t + \mu + d) = x(t + \mu + d)\}_{d \in [\Delta]}]$$



1. Estimation of μ

To find the “best” value of μ we solve this Hamming-distance minimization problem:

$$\min_{\mu \in \{-T, \dots, T\}} \frac{\sum_{(\tau_x, \tau_y) \in \mathcal{L}_\mu} |y_B(\tau_y) - x_B(\tau_x)|}{|\mathcal{L}_\mu|} + \frac{\lambda_\mu}{\sqrt{|\mathcal{L}_\mu|}},$$

where (x_B, y_B) denotes the binarized version of the data, the set $\mathcal{L}_\mu$ represents all the time indexings of the data set, and the regularizer λ_μ is properly chosen.

2. Estimation of Δ

Set $\mathcal{X}_\Delta^t := \{X(t + d)\}_{d \in \Delta}$. Using the binarized sliced detrended data $(\tilde{x}_D(t), y_D(t))$, the “best” value of Δ is selected as the AIC-based minimizer of $(-2 \ln L + 4\Delta + 3)$, with

$$\begin{aligned} & \ln L \\ &= \sum_{t=1}^T Y(t) \ln(\Pr[Y(t) \neq 0 \mid \mathcal{X}_\Delta^t; \pi]) + (1 \\ & \quad - Y(t)) \ln(1 - \Pr[Y(t) \neq 0 \mid \mathcal{X}_\Delta^t; \pi]), \end{aligned}$$

where π is estimated via sample cross-correlation.

3. Estimation of the log-likelihood and posterior error $E(t)$

by applying the Bayes theorem.

3. Association of radar and camera traffic counts

FOR each sensor n , after estimating μ :

1. Build a cost matrix for each vehicle v detected at time t ($C(v, t)$), considering:

- temporal consistency (associations on the same instant are weighted more)
 - minimization of posterior detection error $E(t)$
 - vehicle-class mismatch penalty
2. Apply feasibility constraints:
- admissible time window $\pm \Delta$
 - one-to-one matching consistency
 - penalization of conflicting assignments

t	$X(t)$	$C(v_1, 1)$	$Y(t)$
1	1		1
2	0		0
3	2		0
4	1		2
5	0		3
6	3		0
7	0		0
...	...		...
$\mathbb{Z}$	1		1

The minimum-cost path corresponding to the optimal sensor-oracle matching



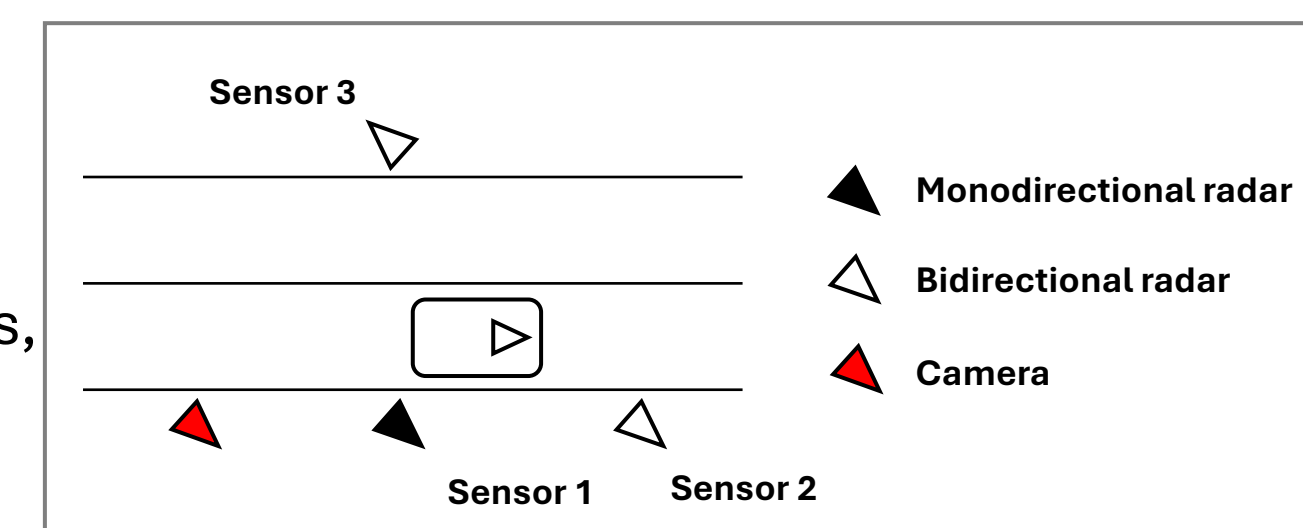
Shortest-path optimization problem on a directed acyclic graph, solved using Dynamic Programming

4. Experimental results

Case study

3 radar sensors and one camera.

Data set: 7200 x 4 data points, sampling time = 1 second.



Estimated first and second order association statistics of the HHMM:

$$\begin{aligned} \mu_1 &= 4, & \mu_2 &= 12, & \mu_3 &= 6 \\ \Delta_1 &= 1, & \Delta_2 &= 3, & \Delta_3 &= 2 \end{aligned}$$

Average sensor posterior errors [%]:

$$\bar{E}_1 = 19.04\%, \quad \bar{E}_2 = 35.73\%, \quad \bar{E}_3 = 27.93\%$$

Total number of camera detection (oracle): 539

Effective number of detections: sensor 1: 498; sensor 2: 510; sensor 3: 501

Vehicle class temporal mismatches:

- sensor 1: 39 mismatches out of 498 association (7.83%)
- sensor 2: 46 mismatches out of 510 association (9.02%)
- sensor 3: 144 mismatches out of 501 association (28.74%)

5. Conclusions and future work

A Bayesian framework for traffic-count quality assessment has been presented. The proposed approach combines probabilistic inference with Operations Research techniques based on shortest-path optimization and Dynamic Programming, to estimate:

- (i) posterior sensor error probabilities;
- (ii) optimal radar-camera associations and systematic temporal mismatches.

Experimental results on real traffic data show that the framework effectively quantifies radar sensor reliability, reconstructs sensor-camera associations, and identifies class-detection mismatches.

Future directions: identification of significant sensors; accounting for vehicular classes in the proposed Bayesian framework; apply the methodology to a wider dataset to identify useful associations between radar configuration, monitored volumes and obtained errors.